

Table 4: Generated summaries from NMT with PS. Boldface words are the words copied from the source.

Source #1	china 's tang gonghong set a world record with a clean and jerk lift of ### kilograms to win the women 's over-## kilogram weightlifting title at the asian games on tuesday .
Target #1	china 's tang <unk>,sets world weightlifting record
NMT+PS #1	china 's tang gonghong wins women 's weightlifting weightlifting title at asian games
Source #2	owing to criticism , nbc said on wednesday that it was ending a three-month-old experiment that would have brought the first liquor advertisements onto national broadcast network television .
Target #2	advertising : nbc retreats from liquor commercials
NMT+PS #2	nbc says it is ending a three-month-old experiment
Source #3	a senior trade union official here wednesday called on ghana 's government to be " mindful of the plight " of the ordinary people in the country in its decisions on tax increases .
Target #3	tuc official,on behalf of ordinary ghanaians
NMT+PS #3	ghana 's government urged to be mindful of the plight

vocabulary, we first check if the same word y_t appears in the source sentence. If it is not, we then check if a translated version of the word exists in the source sentence by using a look-up table between the source and the target language. If the word is in the source sentence, we then use the location of the word in the source as the target. Otherwise we check if one of the English senses from the cross-language dictionary of the French word is in the source. If it is in the source sentence, then we use the location of that word as our translation. Otherwise we just use the argmax of l_t as the target.

For switching network d_t , we observed that using a two-layered MLP with noisy-tanh activation (Gulcehre et al., 2016) function with residual connection from the lower layer (He et al., 2015) activation function to the upper hidden layers improves the BLEU score about 1 points over the d_t using ReLU activation function. We initialized the biases of the last sigmoid layer of d_t to -1 such that if d_t becomes more biased toward choosing the shortlist vocabulary at the beginning of the training. We renormalize the gradients if the norm of the gradients exceed 1 (Pascanu et al., 2012).

Table 5: Europarl Dataset (EN-FR)

	BLEU-4
NMT	20.19
NMT + PS	23.76

In Table 5, we provided the result of NMT with pointer softmax and we observe about 3.6 BLEU score improvement over our baseline.

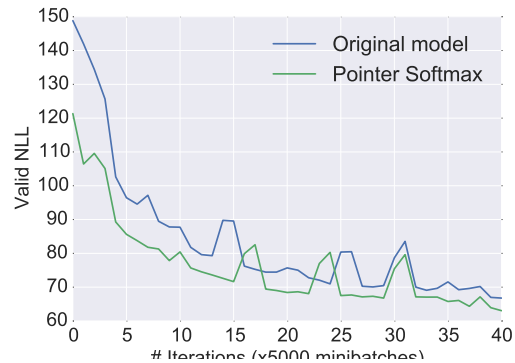


Figure 4: A comparison of the validation learning-curves of the same NMT model trained with pointer softmax and the regular softmax layer. As can be seen from the figures, the model trained with pointer softmax converges faster than the regular softmax layer. Switching network for pointer softmax in this Figure uses ReLU activation function.

In Figure 4, we show the validation curves of the NMT model with attention and the NMT model with shortlist-softmax layer. Pointer softmax converges faster in terms of mini-batch updates and achieves a lower validation negative-log-likelihood (NLL) (63.91) after 200k updates over the Europarl dataset than the NMT